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MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING

Academic Year: 2022-2023

Program: BTech Stream ;CSBS

Year: I Semester: II

Subject: Data Structures and Algorithms

Time: 3 hrs (11:00am to 2:00pm)

Date: 17 /05 /2023.

No. of Pages:2

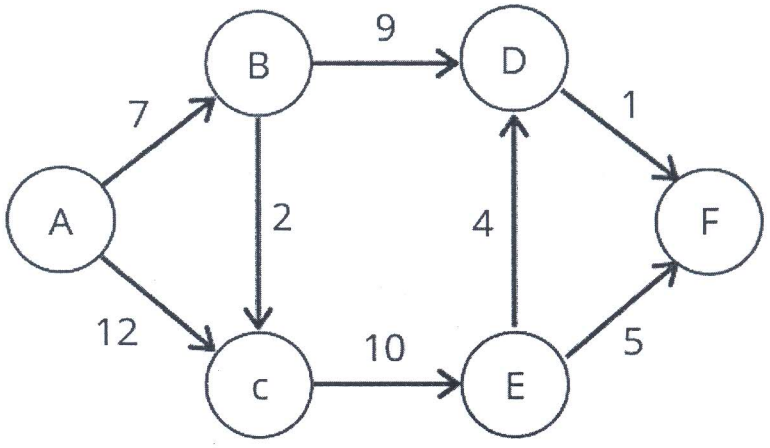
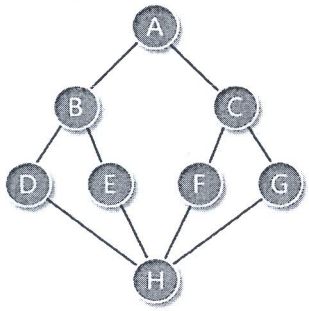
Marks: 100

Re-Examination (Batch 2020-21, 21-22)

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. 1 is compulsory.
2) Out of remaining questions, attempt any 4 questions.
3) **In all 5 questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

Q1 CO-1; BL-2	a.	Explain different asymptotic notations to compute the complexity	[5]
CO- 2; BL- 2	b.	State any one application of stack, queue, trees, linked list and graph data structures.	[5]
CO- 3; BL-2	c.	What are the two ways of representing trees in the memory? Which one do you prefer and why?	[5]
CO- 4; BL-3	d.	Write the algorithm for performing binary search on the sorted array.	[5]
Q2 CO-3; BL-4	a	Create a binary search tree using the following data elements 45, 39, 56, 12, 34, 78, 32, 10, 89, 54, 67, 81	[10]
CO-2 BL-3	b	Write an algorithm to implement insertion and deletion function of queue data structure.	[10]
Q3 CO-3 BL- 3	a	Following graph represent cities in a state. Apply Dijkstra's algorithm to find the optimal path between source city A to all other cities.	[10]

			
CO-3; BL-2	b	Discuss the concept of quick sort with an example	[10]
Q4 CO-2; BL-3	a	Write a program to implement following functions of linked list i. Insert a new node at the beginning of the list. ii. Delete a node at the beginning of the lisr	[10]
CO-4; BL-3	b	What is hashing? Explain the concept of Linear probing. Give applications of hashing	[10]
Q5 CO-3; BL-4	a	Write short note on a) file organization b) expression tree	[10]
CO-2; BL-3	b	Write an algorithm for push and pop operation of Stacks	[10]
Q6 CO-3 BL-3	a	Perform Breadth first search traversal and write an algorithm for the same 	[10]
CO-4; BL-2	b	Consider an array of integers given below. Sort the given values using insertion sort. 39, 9, 45, 63, 18, 81, 108, 54, 72, 36	[10]